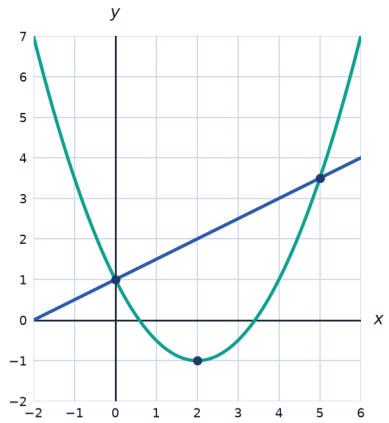
Graphs of linear inequalities (Extended)

Core Concepts: Graphs of linear inequalities (Extended)

1 For the model $f(x) = 6x + 4$ in graphs of linear inequalities (extended), evaluate $f(2)$.



a. 16

b. 22

c. 12

d. 20

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for graphs of linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 4$ in graphs of linear inequalities (extended), evaluate $f(3)$.

a. 22

b. 28

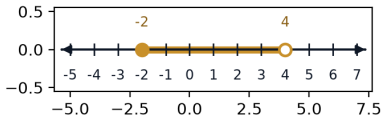
c. 18

d. 26

5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$



c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

6 A data set recorded for graphs of linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 6x + 4$ in graphs of linear inequalities (extended), evaluate $f(4)$.

a. 28

b. 34

c. 24

d. 32

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8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

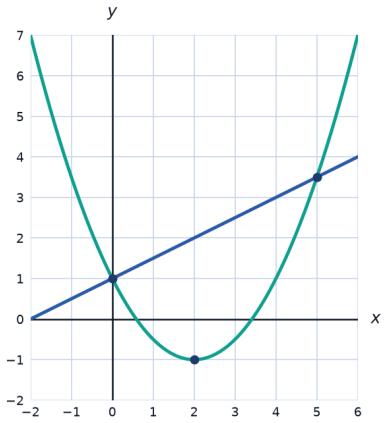
a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

9 A data set recorded for graphs of linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?



a. 15.5

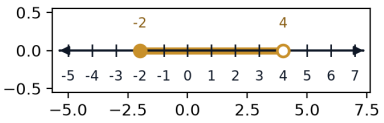
b. 16.5

c. 17.5

d. 19.5

Exam-Style Problem Solving & Modeling

10 For the model $f(x) = 6x + 4$ in graphs of linear inequalities (extended), evaluate $f(5)$.



a. 34

b. 40

c. 30

d. 38

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11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for graphs of linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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